Oleksandr Shchur

oleks.shchur@gmail.com Github Google Scholar +49 174 7515763

Research interests

The main topic of my research is machine learning for temporal data and graphs. Specifically, I'm interested in temporal point processes (probabilistic models for continuous-time event data), generative models, time series forecasting and graph neural networks.

Education

Ph.D. in Computer Science — since 2017
Technical University of Munich, Germany
Advisor: Prof. Stephan Günnemann

M.Sc. in Computational Science and Engineering — 2013 – 2016
Technical University of Munich, Germany
GPA: 1.9 (German grading system)

B.Sc. in Aerospace Engineering — 2009 – 2013
National Aviation University, Kyiv, Ukraine
GPA: 86/100 (Ukrainian grading system)

Work experience

Applied science intern — 2020 – 2021
Machine Learning & Forecasting team, AWS AI, Berlin, Germany
- Added temporal point process support to the GluonTS forecasting library
- Developed a new approach for anomaly detection in event data
- Wrote a survey paper on neural temporal point processes

Scientific employee — since 2017
Technical University of Munich, Germany
- Various research projects on machine learning for temporal and graph data
- Giving lectures & tutorials (Machine Learning, ML for Graphs and Sequential Data)
- Supervised 12 thesis projects and 3 guided research projects

Student research assistant — 2016
Technical University of Munich, Germany
- Implementation and evaluation of new algorithms for community detection in graphs

Skills

Programming languages & frameworks
- Python, PyTorch, TensorFlow, Python Scientific Stack, Git, Bash

Languages
- English (full professional proficiency), Russian (native), Ukrainian (native), German (B2)

Publications

Neural Temporal Point Processes: A Review 2021
Shchur O., Türkmen A. C., Januschowski T., Günnemann S.
International Joint Conference on Artificial Intelligence (IJCAI)
Summary: A survey of neural TPP models and their applications

Fast and Flexible Temporal Point Processes with Triangular Maps 2020
Shchur O., Gao N., Biloš M., Günnemann S.
Advances in Neural Information Processing Systems (NeurIPS)
(Oral presentation, top 1% of submitted works)
Summary: A new way of defining TPPs using non-autoregressive transformations
Code: github.com/shchur/triangular-tpp

Intensity-free Learning of Temporal Point Processes
Shchur O., Biloš M.*, Günnemann S.*
International Conference on Learning Representations (ICLR)
(Spotlight presentation, top 6% of submitted works)
Summary: More efficient TPP models using tools from neural density estimation
Code: github.com/shchur/ifl-tpp

Overlapping Community Detection with Graph Neural Networks 2019
Shchur O., Günnemann S.
Deep Learning on Graphs Workshop, KDD
Summary: A new GNN architecture for scalable detection of overlapping communities
Code: github.com/shchur/overlapping-community-detection

Dual-Primal Graph Convolutional Networks
Monti F., Shchur O., Bojchevski A., Litany O., Günnemann S., Bronstein M.
Graph Embedding and Mining Workshop, ECML–PKDD
Summary: Improve graph neural networks by performing convolutions on the dual graph

Pitfalls of Graph Neural Network Evaluation 2018
Shchur O., Mumme M.*, Bojchevski A., Günnemann S.*
Relational Representation Learning Workshop, NeurIPS
Summary: Better evaluation strategies for GNNs for the node classification task
Code: github.com/shchur/gnn-benchmark

Anomaly Detection in Car-Booking Graphs
Shchur O., Bojchevski A., Farghal M., Günnemann S., Saber Y.
Workshop on Data-driven Intelligent Transportation, ICDM
Summary: Detecting fraud in ride-sharing systems using tensor decomposition

NetGAN: Generating Graphs via Random Walks
Bojchevski A., Shchur O.*, Zügner D.*, Günnemann S.*
International Conference on Machine Learning (ICML)
Summary: An implicit deep generative model for graphs based on Wasserstein GAN

* equal contribution

Scientific community service

(External) reviewer
- ICML, NeurIPS, KDD, AAAI, ICDM, ECML-PKDD